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RESEARCH ARTICLE

Brain Pathology

Gliovascular alterations in sporadic and familial Alzheimer's disease: *APOE3* Christchurch homozygote glioprotection

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Abstract

In response to brain insults, astrocytes become reactive, promoting protection and tissue repair. However, astroglial reactivity is typical of brain pathologies, including Alzheimer's disease (AD). Considering the heterogeneity of the reactive response, the role of astrocytes in the course of different forms of AD has been underestimated. Colombia has the largest human group known to have familial AD (FAD). This group carries the autosomal dominant and fully penetrant mutation E280A in *PSEN1*, which causes early-onset AD. Recently, our group identified an E280A carrier who did not develop FAD. The individual was homozygous for the Christchurch mutation R136S in *APOE3* (APOEch). Remarkably, APOE is the main genetic risk factor for developing sporadic AD (SAD) and most of cerebral ApoE is produced by astroglia. Here, we characterized astrocyte properties related to reactivity, glutamate homeostasis, and structural integrity of the gliovascular unit (GVU), as factors that could underlie the pathogenesis or protection of AD. Specifically, through histological and 3D microscopy analyses of postmortem samples, we briefly describe the histopathology and cytoarchitecture of the frontal cortex of SAD, FAD, and APOEch, and demonstrate that, while astrodegeneration and vascular deterioration are prominent in SAD, FAD is characterized by hyperreactive-like glia, and APOEch displays the mildest astrocytic and vascular alterations despite having the highest burden of Aβ. Notably, astroglial, gliovascular, and vascular disturbances, as well as brain cell death, correlate with the specific astrocytic phenotypes identified in each condition. This study provides new insights into the potential relevance of the gliovasculature in the development and protection of AD. To our knowledge, this is the first study assessing the components of the GVU in human samples of SAD, FAD, and APOEch.

KEYWORDS
APOEch, astrocytes, familial Alzheimer's disease, gliovascular unit, microglia, sporadic Alzheimer's disease

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1 | INTRODUCTION

Astrocytes, microglia, and endothelial cells establish tightly articulated physical and molecular interactions [1–4] in a functionally integrated structure known as the gliovascular unit (GVU) [5] through which many of the properties of the brain are maintained [3, 4]. Astrocytic processes surround most of the vascular surface, regulating the blood–brain barrier (BBB) permeability [6–9]—by modulating tight junction proteins such as claudin-5 [10], or endothelial proteins such vimentin, which is associated with the stability of cerebral vasculature [11]—and integrate neuronal activity with blood flow [1–5, 12–20].

The astroglial cells are the main regulators of cerebral glutamate clearance and metabolism [21–36]. Through the membrane glutamate aspartate transporter (GLAST), the astroglia dynamically remove glutamate from the extracellular milieu, restricting its signaling activity in time and space and allowing for its recycling [25, 30, 33]. Complementarily, the cytoplasmic enzyme glutamine synthetase (GS) catalyzes the condensation of ammonium and glutamate into glutamine [24, 27, 29], which in turn can be reconverted into glutamate for use either as a neurotransmitter or energy substrate [21–27].

Together with microglia, astrocytes constitute the principal defensive system of the CNS [37], responding to numerous damage-related stimuli [6–9] including amyloid β peptides (Aβ) and glutamate overload, in a process known as astrocyte reactivity [15, 37–40]. Through the reactive response, astrocytes aim to restore regional homeostasis gradually modifying their transcriptional profile and adapting their cell morphology and activity [15, 38]. Finely tuned morphological changes are driven by an increase of intermediate filament proteins, such as GFAP and vimentin [15, 38, 41–43], which not only provide mechanical sustenance for the maintenance of astrocyte shape, but also support the functional profile of reactive astrocytes through transcriptional modulation of target genes involved in astroglial and microglial responsiveness [44, 45]. Although reactive astroglia increases trophic support and glutamate uptake, isolates lesioned areas favoring regeneration, activates immune pathways, and promotes BBB repair [38, 43, 45–47], astrocytes can become dysfunctional and acquire cytotoxic potential under prolonged stress [38, 43]. Hence, astrocyte reactivity has frequently been associated with neuroinflammation, unfavorable glial scar formation [7–9, 37, 38, 43, 48, 49], glutamate dyshomeostasis [43, 50, 51], altered coverage of synapses and blood vessels, and BBB breakout [38, 42, 43, 45–47]. The astroglial reactive response is far from a uniform process. Rather, it is a context-specific reaction influenced by brain region, pathology, disease stage, genetics, age, and comorbidities [15, 43, 45, 52], and it is characteristic of multiple brain pathologies, including Alzheimer's disease (AD).

AD is characterized by the aggregation of Aβ and p-tau [53]. The vast majority of AD occurs on an apparently sporadic basis (sporadic AD or SAD) [54]. However, approximately 0.6% of cases present a hereditary form (familial AD or FAD) caused by mutations in *APP*, *PSEN1*, or *PSEN2* that lead to aberrant Aβ overproduction and aggregation [55, 56]. Unlike FAD, SAD is a more complex disorder in terms of the associated triggers. The confluence of environmental and genetic elements is likely to drive the pathology [57]. In particular, vascular risk factors and the allele *APOE4* are the main factors of susceptibility identified to date [5, 58].

Interestingly, Colombia has the largest and most homogeneous family group known to have an early-onset (~35 years of age) form of FAD [54, 59] caused by the autosomal dominant and fully penetrant E280A mutation in *PSEN1* [60, 61], which generally produces a more aggressive course of pathology than SAD [55, 56]. Fascinatingly, our group identified an E280A carrier who did not develop FAD and instead displayed mild cognitive impairment three decades after the expected age of clinical onset [62]. The individual was homozygous for the Christchurch mutation R136S in *APOE3* (APOEch), which has been recognized as its most likely genetic modifier. In contrast to *APOE4*, which has been related to higher risk and younger age of AD onset, *APOE3* has been considered neutral with regard to AD risk [63]. Notably, astrocytes are the major producers of cerebral ApoE [64], which has been reported to play a relevant role in Aβ clearance [63]. Although the mechanisms of protection of ApoE in APOEch have not been identified, it was suggested that these could be associated with a decreased ability to promote Aβ aggregation and neuronal uptake of extracellular p-tau [62]. Full neuropathological characterization of this case showed severe extended Aβ pathology, similar to that observed in other affected E280A carriers, while presenting atypical distribution of p-tau pathology. Frontal cortex of APOEch showed one of the highest accumulations of Aβ compared to other brain regions but significantly lower levels of p-tau [65], making it an interesting region to investigate at the astroglial and gliovascular levels. In fact, APOEch showed upregulation of homeostatic genes in the astrocytes of the frontal cortex and a signature for acute immune response. APOE was among the differentially expressed genes in astrocytes and microglia in APOEch, with an expression gradient from high in frontal cortex to low in occipital cortex, which in turn exhibited a distinctive chronic inflammatory microglial profile and severe p-tau pathology [65]. We have previously reported small vessel disease in SAD, FAD [66, 67], and APOEch [65]. However, the specific changes in astrocytic phenotypes and its coupling to the GVU in SAD, FAD, and APOEch have not been described. Three-dimensional characterization of GFAP, GS, GLAST, and Aquaporin 4 (AQP4) covering the microvasculature is essential to define the gliovascular alterations associated with each type of AD, and a possible protective phenotype in APOEch.

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Although the pathogenesis and pathophysiology of AD are not fully understood [54], Aβ has been postulated to be the direct trigger of the cascade of events that lead to neurodegeneration [66]. However, the linearity of the amyloid hypothesis is controversial as it has been consistently shown that Aβ load does not necessarily correlate with neuronal loss or cognitive decline [67, 68]. Therefore, more integrative hypotheses have suggested that inflammation, excitatory-inhibitory imbalance, or vascular impairment is involved in the series of initial reactions that eventually become chronic and result in irreversible brain dyshomeostasis, neurodegeneration, and AD [42, 66, 69]. We adopted this integrative view and focused on astrocytes as a key element that cuts across those events that are critical for brain health and disease. Previous studies of our group have identified molecular differences between SAD and FAD [55, 56, 70–79] in addition to divergences in the clinical manifestations [55, 56].

Considering the plasticity of the astroglial reactive response and the pathophysiological heterogeneity of SAD, FAD (E280A), and APOEch, in this study, we characterized the astrocytic and gliovascular properties that could play a role in the development or protection of AD. Specifically, we briefly describe the histopathology and cytoarchitecture of the frontal cortex; analyze the distribution and morphology of GFAP, GS, GLAST, and AQP4 astrocytes, and microglia; evaluate the configuration of the GVU and the structural integrity of the capillaries.

2 | MATERIALS AND METHODS

2.1 | Human samples

Middle frontal gyrus cortex samples with low postmortem delays (mean of 4.5 h) from $n = 10$ CNT (controls), $n = 9$ SAD, $n = 9$ FAD (E280A), and $n = 1$ APOEch (E280A), from the Biobank of the University of Antioquia were obtained. Each sample was considered as an independent n. The following sample data were included: sex, age of onset, age of death, disease duration, postmortem index (the time lapse between the patient's death and sampling), CERAD score, Braak staging, Thal phase, NIA-AA criteria, the distribution of TDP-43 and Lewy bodies, cerebral amyloid angiopathy (CAA), APOE genotype, and comorbidities (Table 1 and Table S1).

2.2 | Immunohistochemistry and immunofluorescence

Samples were fixed and cut to 50 μm thick for Nissl staining and immunolabeling. The slices were incubated in primary antibody solution containing mouse anti-Aβ (Invitrogen MA1-91209; 1:500), mouse anti-phospho-tau Ser202-Thr205 (Invitrogen MN1020; 1:500), rabbit anti-vimentin (Abcam ab137321; 1:500), mouse anti-GFAP (Sigma-Aldrich G3893; 1:500), rabbit anti-GFAP (Invitrogen PA5-16291; 1:500), rabbit anti-GS (Invitrogen 701989; 1:500), rabbit anti-GLAST (Invitrogen PA5-19709; 1:500), rabbit anti-IBA1 (Wako 019-19741; 1:500), rabbit anti-AQP4 (Sigma-Aldrich HPA014784; 1:500), or mouse anti-claudin-5 (Invitrogen 35-2500; 1:500). Subsequently, the samples were incubated with corresponding secondary antibody solution containing biotinylated goat anti-mouse (Invitrogen 31800; 1:250) or goat anti-rabbit (Invitrogen B-2770; 1:250) for immunohistochemistry, and goat anti-mouse Alexa Fluor 488 (Invitrogen A-11001; 1:750), goat anti-mouse Alexa Fluor 594 (Invitrogen A-11005; 1:750), goat anti-rabbit Alexa Fluor 488 (Invitrogen A-11008; 1:750), or goat anti-rabbit Alexa Fluor 594 (Invitrogen A-11012; 1:750) for immunofluorescence. The latter also contained human endothelial marker DyLight 649 lectin UEA I (Vector Labs DL-1068; 1:500).

2.3 | Whole slide imaging and unique cell imaging

Nissl and immunohistochemically stained slices were scanned using a NanoZoomer-XR (Hamamatsu) bright-field tissue scanner microscope equipped with a 20× objective (NA 0.75; UPlanSApo; Olympus) and using 40× digital zoom. Three visual fields of 1 mm^2 in area from each slide were processed and analyzed. Cell morphology was assessed using bright-field imaging. Five cells randomly selected per replica (six replicas per sample) were registered in the third layer of the gray matter (300–1000 μm depth) using an Olympus CX35 microscope equipped with a 100× objective (NA 1.25; PlanC N; Olympus) and a Nikon DS-5M camera.

2.4 | Bright-field imaging, processing, and analysis

Images from whole slide and unique cell imaging were processed, segmented, and analyzed using FIJI software (NIH ImageJ) [80]. To calculate the total stained areas, segmentation by intensity thresholding was performed. For cell morphology analysis, at least 30 cells per patient were quantified. Cell somas were manually segmented, while semiautomated processing was used to segment cell processes [81]. Morphological parameters related to complexity (fractal dimension, lacunarity, roughness, density, and compactness), size (area, perimeter, and principal axes), shape (circularity and aspect ratio), and spatial domain (scale and shape features of convex hull, bounding rectangle, and fit ellipse) were analyzed [82]. Image processing of Nissl, Aβ, p-tau, claudin-5, and vimentin were performed in QuPath 0.3.2 software [83]. As the thickness of the layers in the gray matter (GM) was variable, regions of interest (ROIs) of regular areas were normalized and used to calculate Nissl-stained cells/mm^2.

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TABLE 1 Subject demographic and pathological information

ID	Condition	Sex	Age of onset	Age of death	Disease duration	PI	CERAD	Braak	Thal	NIA-AA	Comorbidities
C1	CNT	M	NAP	69	NAP	6.8	A	1	2	A1, B1, C1	Hypertension, chronic venous insufficiency
C2	CNT	M	NAP	61	NAP	5.3	0	0	0	A0, B0, C0	Hypertension, type II diabetes mellitus
C3	CNT	F	NAP	44	NAP	4.4	0	0	0	A0, B0, C0	Depression
C4	CNT	M	NAP	38	NAP	3.4	0	0	0	A0, B0, C0	None
C5	CNT	M	NAP	72	NAP	8.4	0	2	0	A0, B1, C0	Hypertension, coronary heart disease, obesity, chronic obstructive pulmonary disease
C6	CNT	M	NAP	73	NAP	3.5	B	1	2	A1, B1, C2	None
C7	CNT	F	NAP	84	NAP	4.3	B	1	2	A1, B1, C2	Hypertension, type II diabetes mellitus, pancreatitis
C8	CNT	F	NAP	67	NAP	3.8	0	1	0	A0, B1, C0	Hypertension, hypothyroidism, lung cancer
C9	CNT	F	NAP	75	NAP	14.8	0	0	1	A1, B0, C0	Hypertension
C10	CNT	F	NAP	91	NAP	12.5	A	2	2	A1, B1, C1	Hypokalemia, Meniere's vertigo
	Average			*67.4*		*6.7*					
	SEM			*5.2*		*1.3*					
S1	SAD	F	81	94	13	4.0	B	4	5	A3, B2, C2	Hypertension, type II diabetes mellitus, breast cancer
S2	SAD	F	79	90	11	3.3	C	4	4	A3, B2, C3	Hypertension, dyslipidemia
S3	SAD	F	59	73	14	3.5	C	5	4	A3, B3, C3	Dyslipidemia, depression
S4	SAD	F	92	98	6	4.3	A	3	3	A2, B2, C1	Hypertension
S5	SAD	M	52	68	16	5.1	B	5	5	A3, B3, C2	Hypertension, coronary heart disease, dyslipidemia, alcoholism
S6	SAD	F	75	85	10	3.3	B	5	5	A3, B3, C2	Hypertension, congestive heart failure, single hemorrhagic stroke, hypothyroidism
S7	SAD	F	83	86	3	2.8	C	6	5	A3, B3, C3	Hypertension, dyslipidemia, chronic gastritis
S8	SAD	F	83	97	14	4.3	C	6	5	A3, B3, C3	Cerebrovascular disease
S9	SAD	F	75	91	16	3.4	C	6	5	A3, B3, C3	Hypothyroidism, kidney stones
	Average		*75.4*	*86.9*	*11.4*	*3.8*					
	SEM		*4.2*	*3.4*	*1.5*	*0.2*					
F1	FAD (E280A)	F	44	50	6	5.2	C	6	5	A3, B3, C3	None

(Continues)

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TABLE 1 (Continued)

ID	Condition	Sex	Age of onset	Age of death	Disease duration	PI	CERAD	Braak	Thal	NIA-AA	Comorbidities
F2	FAD (E280A)	M	46	62	16	4.3	C	6	5	A3, B3, C3	Hypertension, alcoholism, traumatic brain injury
F3	FAD (E280A)	F	50	63	13	4.9	C	6	5	A3, B3, C3	Chronic venous insufficiency, hypothyroidism
F4	FAD (E280A)	M	49	59	10	5.0	C	6	5	A3, B3, C3	Dyslipidemia, alcoholism
F5	FAD (E280A)	F	51	65	14	3.8	B	5	5	A3, B3, C2	Dyslipidemia, type II diabetes mellitus, hypothyroidism
F6	FAD (E280A)	F	41	56	15	5.0	C	5	5	A3, B3, C3	Hypertension
F7	FAD (E280A)	M	44	57	13	1.8	C	5	5	A3, B3, C3	Traumatic brain injury
F8	FAD (E280A)	F	46	67	21	4.0	B	4	5	A3, B2, C2	Depression
F9	FAD (E280A)	F	49	62	13	4.0	B	4	5	A3, B2, C2	Depression
	Average		*46.7*	*60.1*	*13.4*	*4.2*					
	SEM		*1.1*	*1.7*	*1.4*	*0.3*					
APOEch	APOEch (E280A)	F	72[a]	77	5[a]	3.2	C	6	5	A3, B3, C3	Hypertension, dyslipidemia, metastatic melanoma

Note: Demographic, histopathological, and clinical data of the cohorts, donors, and samples used in this study are shown. The E280A carrier, which was also homozygous for the Christchurch mutation R136S in *APOE3*, is referred to as APOEch throughout the manuscript.
Abbreviations: F, female; M, male; NAP, not applicable; ND, no data available; PI, postmortem index.
[a]The age of onset and disease duration of APOEch refer to a mild cognitive impairment. Disease duration in years and postmortem index in hours are shown.

Manual count of extravascular claudin-5 was performed. Classification of Aβ plaques was done using machine learning (Figure S1). ROIs were manually drawn over plaques and classified by an experienced neuropathologist as "dense" or "diffuse." The training process was performed on every slide.

2.5 | Confocal imaging

Three high-magnification images per slide in the third layer of the GM were obtained. At least 30 fields per sample were quantified. GS-GFAP-UEA I images were acquired using a Perkin Elmer Ultraview RS Spinning Disk Confocal Microscope equipped with a 40× oil-immersion objective (NA 1.3; C-Apochromat; Zeiss). Images of 139.44 × 106.24 μm were acquired with an XY pixel size of 103.75 nm and a distance of 300 nm between Z-sections. Approximately 15 μm thick in each field were registered. GLAST-GFAP-UEA I and AQP4-GFAP-UEA I images were obtained using an Olympus FV1000 confocal scanning microscope equipped with a 60× oil-immersion objective (NA 1.42; PLAPON; Olympus). Images of 105.47 × 105.47 μm in size were obtained with an XY pixel size of 103 and 500 nm between Z-sections. 10.5 μm thick of each field were registered.

2.6 | Confocal imaging, processing, and analysis

Image deconvolution was performed using Huygens Professional software, version 19.10 (Scientific Volume Imaging, the Netherlands, http://svi.nl). Images were segmented through machine learning, processed in Ilastik software version 1.3 [84], and analyzed in FIJI software [80]. Z projections of segmented vessels were used to measure the vascular diameter. For this purpose, internal 3D distance maps were built [85]. The average radius along the whole length of the vessels obtained through skeletonized Z projections was reported. Z projections were also used to quantify vascular gaps, assuming that the number of gaps on the projected plane was similar to those planes that were not projected. External distance maps were employed to characterize the associations of astrocytic perivascular processes with the vessels, excluding astrocytic signals coming from 2 μm from the vessel surface (Figure S6).

2.7 | GVU regulatory gene network analysis

Through single nuclei sequencing, differentially expressed genes were identified in astrocytes of hippocampus,

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occipital, and particularly in the frontal cortex of APOEch [65]. We searched in the literature for genes assigned to GVU regulation [86–88] and performed gene network and enrichment analyses using *metascape* toolset [89].

2.8 | Statistical analysis

The sample size for each experimental group was defined as each patient and analyzed as an independent experiment. Normality was evaluated by the Shapiro–Wilk test. We analyzed parametric univariable data using one-way ANOVA and Tukey's multiple comparison tests. Nonparametric Kruskal–Wallis test and Dunn's multiple comparison test were performed when either of the datasets was not normally distributed. Pairwise comparisons were also performed with Student's unpaired two-tailed *t* test when both datasets were normally distributed. We processed all of the groups in parallel to reduce interassay variation. Although FAD and APOEch are shown together in the graphs, they were analyzed separately in all of the experiments. The analyses were performed in GraphPad Prism software, version 8.0. At least five replicates were measured for each sample in all experiments. Data are presented as the mean ± standard error of the mean (SEM) from at least four independent experiments. The results were considered significant from $p < 0.05$.

Complete description of materials and methods can be found in Appendix S1.

3 | RESULTS

3.1 | Cellular deterioration of frontal cortical layers I–III is associated with Aβ and p-tau in SAD and FAD but not in APOEch

In accordance with what is known for late- and early-onset AD [56], the duration of the disease was similar between SAD and FAD (Table 1 and Table S1). In contrast, despite carrying the E280A mutation, the APOEch donor developed only mild cognitive impairment at 72 years of age and died at 77 from causes unrelated to this condition. Moreover, APOEch presented the highest histopathological scores (CERAD, Braak, Thal, and NIA-AA), followed by FAD and then SAD.

We compared the histopathology and cytoarchitecture of the gray matter (GM) of the frontal cortex in SAD, FAD, and APOEch. The highest percentage of Aβ area was displayed by APOEch, followed by FAD, and then SAD (Figure 1A,B). Notably, Aβ was especially abundant within the first millimeter of depth of the GM in SAD and FAD, while in APOEch, it was widely spread throughout the GM. In addition, the deposits of Aβ displayed a more condensed appearance in SAD, followed by FAD and APOEch (Figure S1). The latter exhibited the most filamentous staining pattern. Similarly, the percentage of p-tau was higher in FAD than in SAD, as previously reported [77], but the lowest levels were found in APOEch which was similar to CNT (Figure 1C,D). Consistently with the presence of Aβ and p-tau, the number of cells above and below 1 mm depth—which approximately corresponds to neuronal layers I–III and IV–VI of the GM, respectively [90]—revealed a significant cell density decrease in the upper layers in SAD and FAD, while only SAD displayed a reduction in cell density in the lower layers (Figure 1E, F). APOEch presented the highest cell density among the E280A group in layers I–III, and did not show remarkable differences within the group in the lower layers. As an approximation to determine the type of cells lost in the upper layers of the GM, we filtered by cell size considering cell bodies larger than 120 μm^2 as typical pyramidal-like cells, and smaller cell bodies as nonpyramidal-like cells. While the cell density decrease in layers I–III in SAD and FAD was caused by the loss of both nonpyramidal-like and pyramidal-like cells, the density of pyramidal-like cells in APOEch was comparable to that in CNT (Figure 1G). Together, our results suggest that layers I–III of the GM are particularly affected in the frontal cortex of AD in terms of both the presence Aβ and p-tau and cellular density. In this respect, APOEch presented relatively low cytoarchitectural disturbances and low p-tau, despite exhibiting the highest levels of Aβ.

3.2 | Differential distribution of reactive-like astrocytes in the frontal cortex of AD and APOEch

Considering the deterioration of layers I–III and the contrasting findings in APOEch, we evaluated the abundance, distribution, and morphology of vimentin and GFAP astrocytes to estimate the state of astroglial reactivity in this region. As previously described for healthy brains [91], vimentin astrocytes were almost restricted to layer II in CNT, and their distribution in APOEch was remarkably similar to that, accumulating 80% of cells at around 500 μm in depth (Figure 2A,C). Meanwhile, vimentin astrocytes were spread throughout the GM in SAD and FAD such that at 500 μm depth, only 18% and 35% of the cells were found, respectively. Regarding GFAP, the gold-standard marker of astrocyte reactivity, it displayed highly dissimilar staining patterns between conditions (Figure 2B and Figure S2A). On the one hand, the percentage of GFAP area was the highest in FAD, followed by SAD, and the lowest in APOEch which was strikingly similar to CNT (Figure 2D). Notably, astrovascular associations and highly branched astrocytic morphologies were observed in vimentin and GFAP staining in layer II of CNT and APOEch, whereas these findings were much less frequent in AD (Figure 2A,B and Figure S3). Consistent with previous reports [90, 91]

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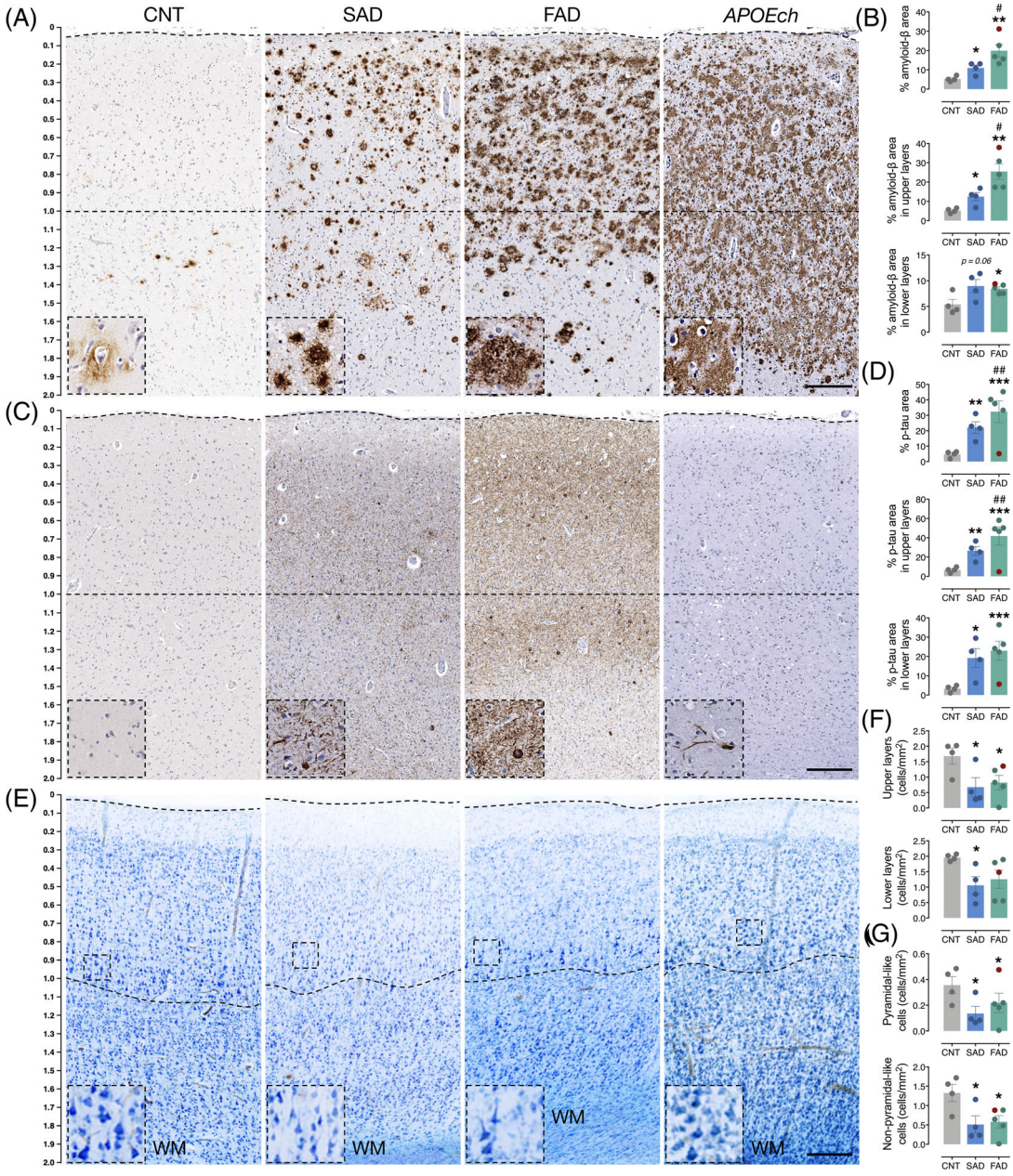


FIGURE 1 Characteristic histopathology and cytoarchitecture of the GM in the frontal cortex of SAD, FAD, and APOEch. (A) Representative images of Aβ immunohistochemistry showing different loading and distribution of Aβ plaques along the GM. Most Aβ is found within the first millimeter depth (indicated by dashed lines) in SAD and FAD. Insets show the characteristic pattern of Aβ aggregation found in each condition. (B) Quantification of Aβ area in different regions of the GM. (C) Characteristic staining pattern of hyperphosphorylated tau showing higher presence of neurofibrillary tangles in FAD than in SAD and scarce in APOEch. Insets show the typical appearance of tau-positive structures for each condition. (D) Quantification of p-tau area in different regions of the GM. Upper and lower layers in (B) and (D) correspond to 0–1 mm and 1–2 mm, respectively. (E) Representative images of Nissl staining showing differential cytoarchitectural features between SAD, FAD, and APOEch. Dashed lines enclose neuronal layers I–III. WM indicates white matter. Insets are magnification of the boxes in layer III highlighting cell density and cell body size. (F) Quantification of cell density in the upper (I–III) and lower (IV–VI) layers of the GM, and (G) of pyramidal-like (soma >120 μm^2) and non-pyramidal-like (soma <120 μm^2) cells in the upper layers. The rulers on the left indicate the depth in millimeters in (A), (C), and (E). Scale bars in (A), (C), (E): 250 μm. Insets in (A), (C), (E): 5× magnification. Data are presented as mean ± SEM from CNT, $n = 5$ (C1, C2, C3, C4, C5); SAD, $n = 4$ (S1, S2, S3, S4); and FAD, $n = 5$ (F2, F4, F5, F6, F7); independent experiments. The red point is the mean of three replicates of APOEch. One-way ANOVA, *t*-test, and Tukey's multiple comparison tests were performed. Comparison with CNT is denoted by asterisk (*); with SAD by number sign (#). $^{*/\#}$ indicates $p < 0.05$, $^{**/\#\#}$ $p < 0.01$, and $^{***/\#\#\#}$ $p < 0.001$

and similar to vimentin, GFAP astrocytes were almost limited to layer II in CNT, while they were abundant throughout layers I–III in AD. Interestingly, APOEch showed appreciably low levels of GFAP beyond layer II. To determine whether the increase in GFAP occurred homogenously throughout layers I–III or whether it was

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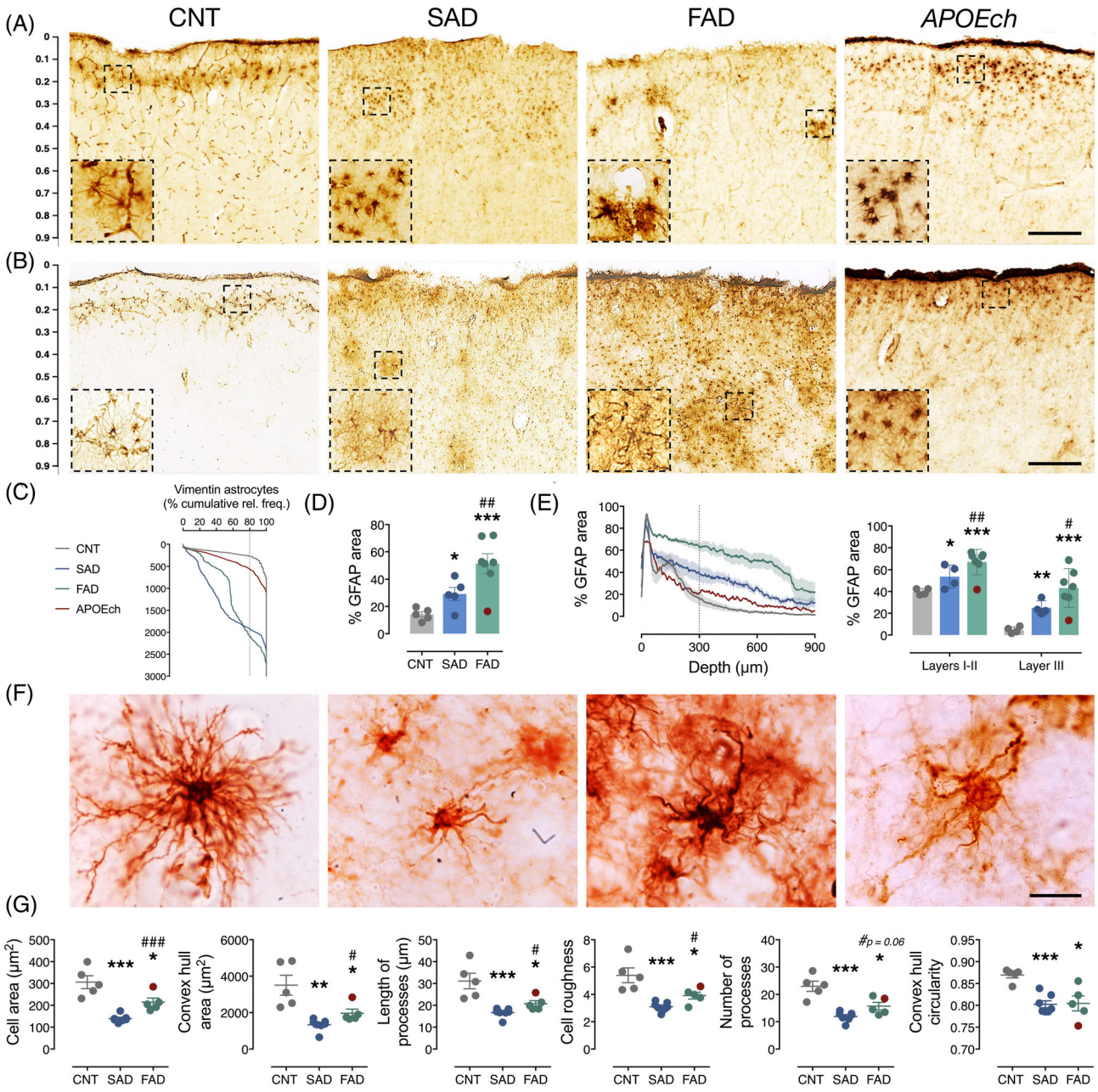


FIGURE 2 Differential abundance, distribution, and morphology of reactive-like astroglia in the frontal cortex of SAD, FAD, and APOEch. (A, B) Representative images of layers I–III of the GM immunostained against (A) vimentin and (B) GFAP. (A) The distribution of vimentin-stained astrocytes is similar in SAD and FAD, while in APOEch it resembles that of CNT. (B) The abundance and distribution of GFAP differ between SAD and FAD, but is similar in APOEch and CNT. Insets highlighting scattered astrocytic clusters in SAD, sharply stained-astrocytes with intertwined processes in FAD, and non-overlapping astrocytes in APOEch. Astrovascular associations under both vimentin and GFAP immunostaining are common in layer II of CNT, but not in AD nor APOEch (insets in A and B). The rulers on the left indicate the depth in millimeters in (A) and (B). (C) Cumulative relative frequency of vimentin astrocytes along the GM. (D) Quantification of percentage of GFAP area in layers I–III. (E) Distribution analysis of GFAP astrocytes along layers I–II (<300 μm depth) and III (>300 μm depth). (F) Representative images of typical GFAP astrocytes found in layer III showing prominent morphological variations between conditions. (G) Analysis of morphometric parameters related to scale (area and length of processes), complexity (roughness and number of processes), and shape (circularity) of GFAP astrocytes and their spatial domains (convex hull). Scale bars in (A) and (B): 250 μm; F: 20 μm. Insets in (A) and (B): 3× magnification. Data are presented as mean ± SEM from CNT, $n = 7$ (C1, C2, C3, C4, C5, C6, C10); SAD, $n = 7$ (S1, S2, S3, S4, S7, S8, S9); and FAD, $n = 6$ (F1, F2, F4, F5, F6, F7); independent experiments. The red point is the mean of three replicates of APOEch. One-way ANOVA, *t*-test, and Tukey's multiple comparison tests were performed. Comparison with CNT is denoted by asterisk (*); with SAD by number sign (#). $^{*/\#}$ indicates $p < 0.05$, $^{**/\#\#}$ $p < 0.01$, and $^{***/\#\#\#}$ $p < 0.001$

associated with a particular layer, we compared the percentage of GFAP area at different depths of the GM. Distribution analysis of GFAP revealed greater differences between conditions beyond 300 μm depth, which approximately corresponds to layer III of GM (Figure 2E). The percentage of GFAP area in FAD was

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meaningfully greater than in CNT and SAD, while APOEch displayed comparable values to those of CNT. Of note, APOEch displayed more astrocytes in the frontal cortex than in other brain regions (Figure S4A,B). Considered together, these results showed a differential abundance and distribution of the reactivity-related markers vimentin and a GFAP in the frontal cortex of human SAD, FAD, and APOEch.

3.3 | Astrocytes are smaller and less complex in SAD, hyperreactive-like in FAD, and similar to those of CNT in APOEch

Considering the differential abundance of GFAP in AD and APOEch, and as GFAP influences astrocyte morphology [45, 91–93], we analyzed morphometric descriptors related to the scale, complexity, and shape of both the astrocytes themselves and their spatial domains. Morphological differences in GFAP astrocyte morphology were evident (Figure 2F). Specifically, the largest (*area* and *length of processes*) and most complex astrocytes (*roughness* and *number of processes*) were those of CNT, followed by FAD, and meaningfully less complex and smaller in SAD (Figure 2G). Surprisingly, APOEch presented the largest and most complex astrocytes within the E280A group. Likewise, astrocytic spatial domains were the largest in CNT, followed by APOEch, FAD, and finally SAD. In addition, CNT astrocytes exhibited the most radial configuration (*convex hull circularity*), followed by FAD and SAD, while APOEch presented the most stretched. Notably, APOEch displays larger astrocytes in the frontal cortex than in other brain regions (Figure S4A,B). A complementary set of cell shape descriptors is shown in Figure S5A. Altogether, these results show that the morphology of human astrocytes in layer III of the GM is more compromised in the frontal cortex of SAD than in FAD. Outstandingly, astrocyte morphology in APOEch resembles that of CNT.

3.4 | Differential abundance of reactive-like microglia in the frontal cortex of SAD, FAD, and APOEch

Given the tight functional relationship between astrocytes and microglia, we examined the abundance, distribution, and morphology of microglial cells in the upper layers of the GM by IBA1 immunostaining. Microglial cells exhibited a homogeneous distribution throughout layers I–III in CNT, while clusters and microglial-devoid areas were recurrent in SAD and FAD (Figure 3A). Similarly, SAD and FAD displayed greater percentages of IBA1 area than CNT, while APOEch exhibited the lowest (Figure 3B). Similar to GFAP, distribution profiles of IBA1 showed greater differences between conditions beyond 300 μm depth. In particular, IBA1 in layer III was higher in FAD than in SAD, while APOEch showed similar levels to CNT (Figure 3C). Concerning morphology, microglial cells of AD tended to be smaller (*area* and *length of processes*) than in CNT and exhibited ameboid characteristics (*density* and *circularity*) (Figure 3D, E). Nonetheless, no differences in microglial morphology were found between SAD and FAD. Intriguingly, the microglia of APOEch presented the smallest and simplest morphology, and exhibited less reactive-like characteristics in the frontal cortex than in other brain regions (Figure S4C,D). Together, our results suggest a differential microglial increase in the GM of AD and APOEch, and a morphological shift toward an ameboid morphology in the three conditions.

3.5 | The expression of astrocytic glutamate-regulatory proteins is differently disturbed in the frontal cortex of SAD, FAD, and APOEch

As astroglial reactivity impacts astrocyte-dependent glutamate homeostasis [36, 94–97], we assessed the expression patterns of the astrocytic glutamate-regulatory proteins GS and GLAST in layers I–III of the GM. The percentage of GS area increased in SAD, whereas nonsignificant differences were identified between CNT, FAD, and APOEch (Figure 4A,B and Figure S2B,C). Contrarily, the percentage of GLAST area was lower in FAD than in SAD, but APOEch displayed the highest levels within the E280A group. Considering that astrocytes are organized into distinct functional-associated layers throughout the GM [98–101], we compared the distribution of GS and GLAST along layers I–III. As previously described [102], GS astrocytes were present from layer III onward in CNT, whereas in SAD and FAD increased from layer I (Figure 4C). APOEch instead, showed a similar distribution pattern to that of CNT. Comparably, GLAST exhibited a diffuse expression pattern from layer III onward in CNT, while in SAD and FAD, it was homogenously distributed throughout all of the upper layers presenting a distinctive aggregated pattern in layer III that was also detected in APOEch. Quantifications revealed that SAD exhibits the highest levels of GS and GLAST in layers I–II, and the highest of GS in layer III. Meanwhile, FAD and APOEch show levels of GLAST similar to CNT in layers I–II, but significantly decreased in layer III. Although morphological differences in GS were not as evident as in GFAP, astrocytes did display greater size and complexity in both types of AD compared to CNT, while those of APOEch were even larger (Figure 4D). A complementary set of cell morphometric descriptors is shown in Figure S5B. Moreover, although GLAST exhibited mostly an aggregated staining pattern in layer III of AD and APOEch as shown by *spreading factor*, the aggregates were more abundant in FAD than in any other condition (Figure 4E). In summary, we found that GS and GLAST

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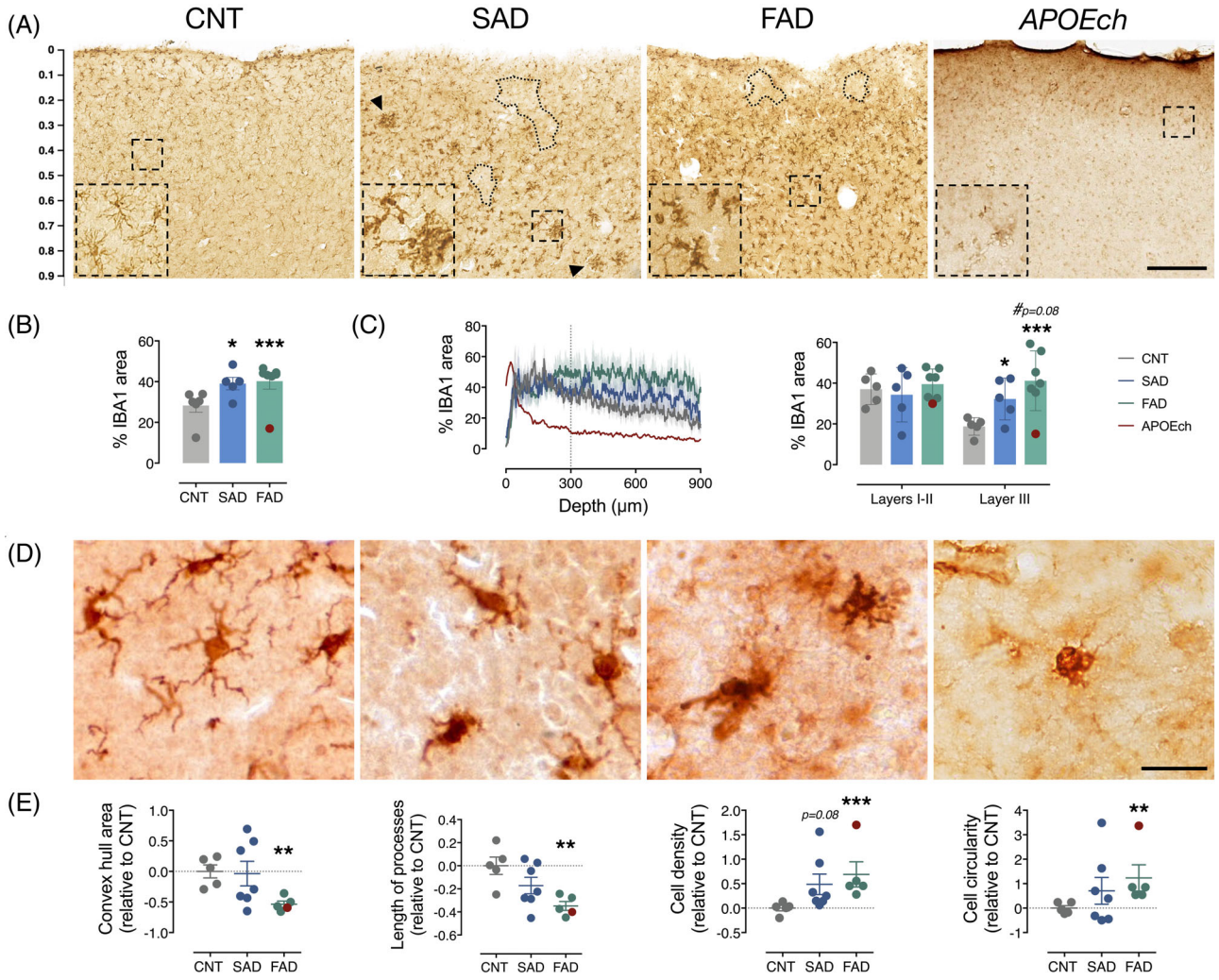


FIGURE 3 Abundance, distribution, and morphology of microglia in the frontal cortex of SAD, FAD, and APOEch. (A) Representative images of layers I–III of the GM immunostained against IBA1 showing regular distribution of microglial cells in CNT, microglial devoid areas (dashed polygons) and cell clusters (arrowheads) in SAD and FAD, and atrophied-like microglia in APOEch. Insets highlight the typical microglial distribution pattern of each condition. The ruler on the left indicates the depth in millimeters. (B) Quantification of percentage of IBA1 area in layers I–III. (C) Distribution analysis of the microglia along layers I–II (<300 μm depth) and III (>300 μm depth). (D) Representative images of characteristic microglial cells found in layer III. (E) Analysis of morphometric parameters related to scale (convex hull area and length of processes), complexity (density), and shape (circularity) of the microglia. Scale bars in (A): 250 μm; (D): 20 μm. Insets in (A): 3× magnification. Data are presented as mean ± SEM from CNT, $n = 6$ (C1, C2, C7, C8, C9, C10); SAD, $n = 8$ (S1, S2, S3, S5, S6, S7, S8, S9); and FAD, $n = 6$ (F1, F2, F3, F4, F8, F9); independent experiments. The red point is the mean of three replicates of APOEch. One-way ANOVA, *t*-test, and Tukey's multiple comparison tests were performed. Comparison with CNT is denoted by asterisk (*); with SAD by number sign (#). $*^{/\#}$ indicates $p < 0.05$, $**^{/\#\#}$ $p < 0.01$, and $***^{/\#\#\#}$ $p < 0.001$

are altered in AD and APOEch. However, subtle differences were identified between conditions mainly in terms of abundance. APOEch presented a mixed expression pattern of GS and GLAST with characteristics from both CNT and AD.

3.6 | The structural configuration of the GVU is differently altered in AD, and to a lesser extent in APOEch

The integrity of the GVU relies on astrocyte morphological and functional stability [2, 3, 17]. In fact, astroglial reactivity [1], astrocyte-mediated inflammation [50], and glutamate dyshomeostasis [51] are associated with vascular deterioration and BBB breakdown. To characterize the configuration of the GVU, we analyzed the astrocytic coverage of the vasculature in layer III of the GM through confocal microscopy and 3D reconstruction. Triple immunolabeling of GS-GFAP-UEA I and GLAST-GFAP-UEA I showed different magnitudes of vascular coverage and distinct predominant astrocytic-perivascular identities in each condition (Figure 5A–B′). To determine the percentage of the vascular surface covered by astrocytic processes, we empirically identified 2 μm as the distance at which perivascular processes

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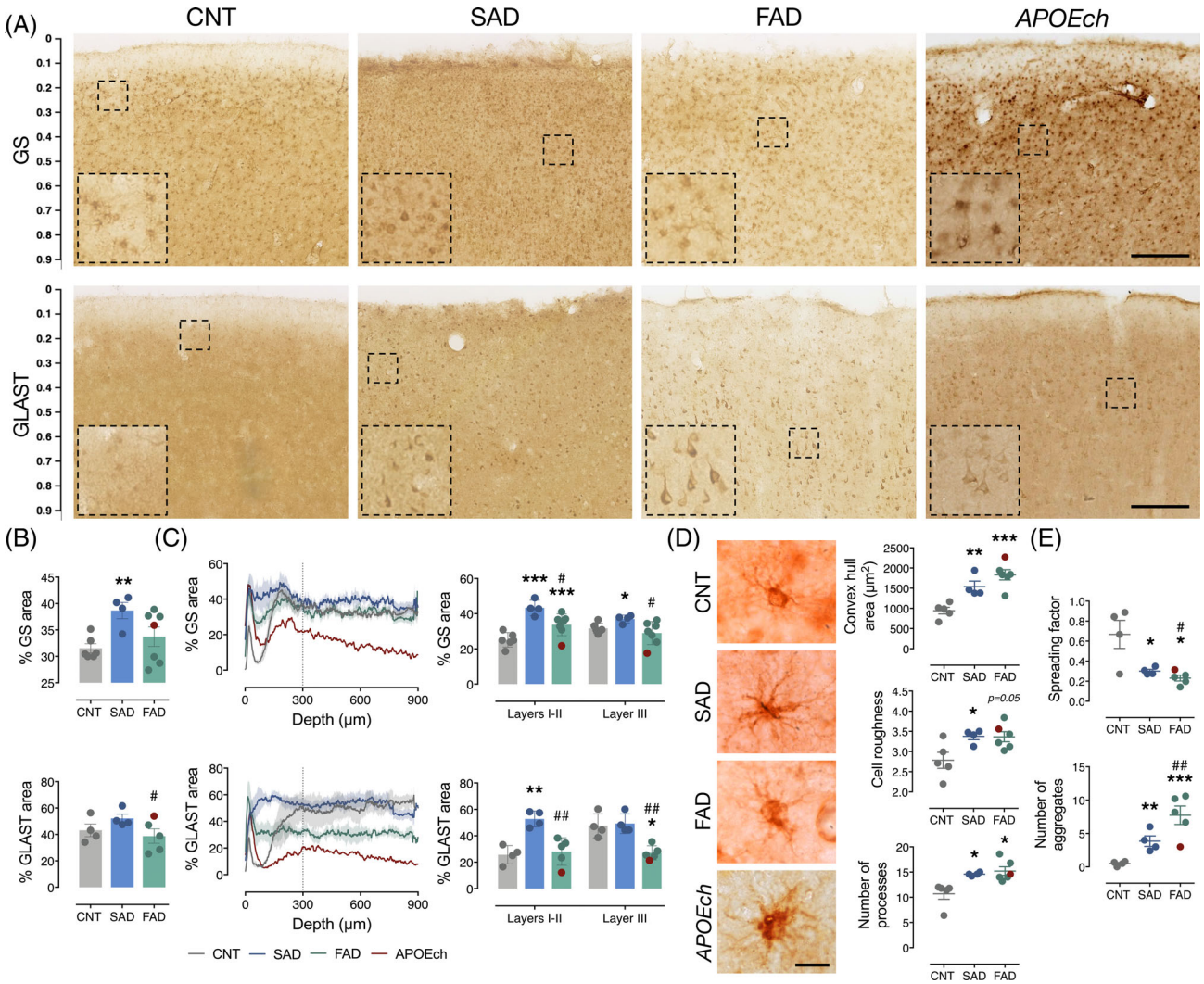


FIGURE 4 Differential abundance and distribution of glutamate-regulatory astrocytic proteins in the frontal cortex of SAD, FAD, and APOEch. (A) Representative images of layers I–III of the GM immunostained against GS and GLAST. Note the increase of GS in the upper edge of the GM and the condensed staining pattern of GLAST in AD and APOEch. Insets highlight characteristic staining pattern of GS and GLAST of each condition. The rulers on the left indicate the depth in millimeters. (B) Quantification of percentage of GS and GLAST area in layers I–III. (C) Distribution analysis of GS and GLAST along layers I–II (<300 μm depth) and III (>300 μm depth). (D) Representative images of characteristic GS astrocytes found in layer III, and analysis of some of their morphometric parameters related to scale (convex hull area) and complexity (roughness and number of processes). (E) Analysis of GLAST staining pattern showing dispersion (spreading factor) and number of aggregates. Scale bars in (A): 250 μm; (D): 20 μm. Insets in (A): 3× magnification. Data are presented as mean ± SEM from CNT, $n = 6$ (C1, C2, C3, C4, C5, C6); SAD, $n = 4$ (S1, S2, S3, S4); and FAD, $n = 7$ (F1, F2, F3, F4, F5, F6, F7); independent experiments. The red point is the mean of three replicates of APOEch. One-way ANOVA, *t*-test, and Tukey's multiple comparison tests were performed. Comparison with CNT is denoted by asterisk (*); with SAD by number sign (#). $^{*/\#}$ indicates $p < 0.05$, $^{**/\#\#}$ $p < 0.01$, and $^{***/\#\#\#}$ $p < 0.001$

encircle the abluminal surface of the vessels (Figure S6). In particular, we found that FAD exhibits the highest percentage of vascular surface surrounded by GFAP processes, whereas APOEch showed the lowest within the E280A group, and SAD is similar to CNT (Figure 5C). Contrastingly, the vascular area covered by GLAST processes or by GFAP-GLAST was meaningfully greater in SAD than in CNT, FAD, and APOEch. In addition, the percentage of the vascular surface covered by astrocytes of any type (sum of GFAP, GS, and GLAST coverage percentages) tends to increase in FAD and decrease in SAD, but remains similar to CNT in APOEch. Surprisingly, although the vessels in SAD appeared to be highly covered by GS, the quantification did not reveal any difference from the rest of the conditions. Therefore, we presumed a change in the proportion of the astrocytic markers associated with the vessels. We found that GFAP decreases while GS and GLAST increase in the perivasculature of SAD in relation with complementary markers (Figure 5D). In contrast, a decrease in the proportion of double-positive GFAP-GLAST and a trend toward lower GS were identified in the perivasculature of

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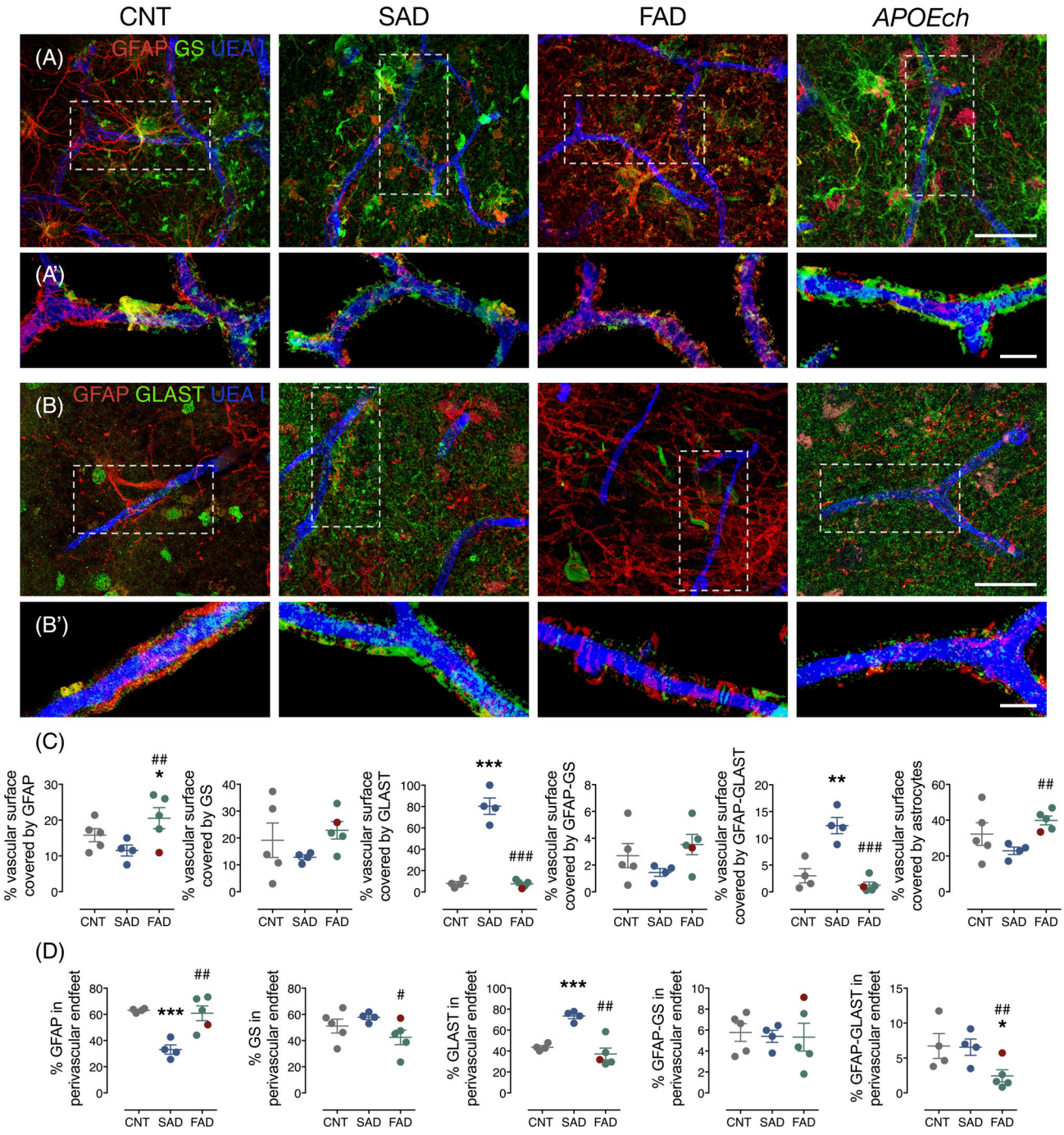


FIGURE 5 The structural configuration of the GVU is differently altered in the frontal cortex of SAD, FAD, and APOEch. (A-B′) Representative confocal images in layer III of the GM showing (A, A′) GFAP-GS and (B, B′) GFAP-GLAST astrocytes near the endothelium labeled with UEA I lectin. (A, B) Maximum intensity z-projections of confocal images. (A′, B′) 2× magnified 3D reconstructions of the boxes in (A) and (B) showing the perivascular astrocytic processes present between 0 and 2 μm away from the vascular surface. Vertical boxes in (A) and (B) are flipped 90° to the right in (A′) and (B′). (C, D) Quantification of (C) the percentage of vascular surface covered by each type of astrocytic processes; and of (D) the predominant astrocytic identity present in the astrocytic perivasculature. Scale bars in (A), (B): 25 μm; (A′), (B′): 10 μm. Data are presented as mean ± SEM from CNT, $n = 6$ (C1, C2, C3, C4, C5, C6); SAD, $n = 4$ (S1, S2, S3, S4); and FAD, $n = 5$ (F1, F2, F4, F5, F6); independent experiments. The red point is the mean of three replicates of APOEch. One-way ANOVA, *t*-test, and Tukey's multiple comparison tests were performed. Comparison with CNT is denoted by asterisk (*); with SAD by number sign (#). $^{*/\#}$ indicates $p < 0.05$, $^{**/\#\#}$ $p < 0.01$, and $^{***/\#\#\#}$ $p < 0.001$

FAD. APOEch only differed from CNT at increasing the proportion of perivascular GFAP-GS. Together, our results show that while FAD proportionally increases the vascular coverage of GFAP, GS, and GLAST astrocytic processes, there is shift in the balance of GFAP and GLAST in SAD.

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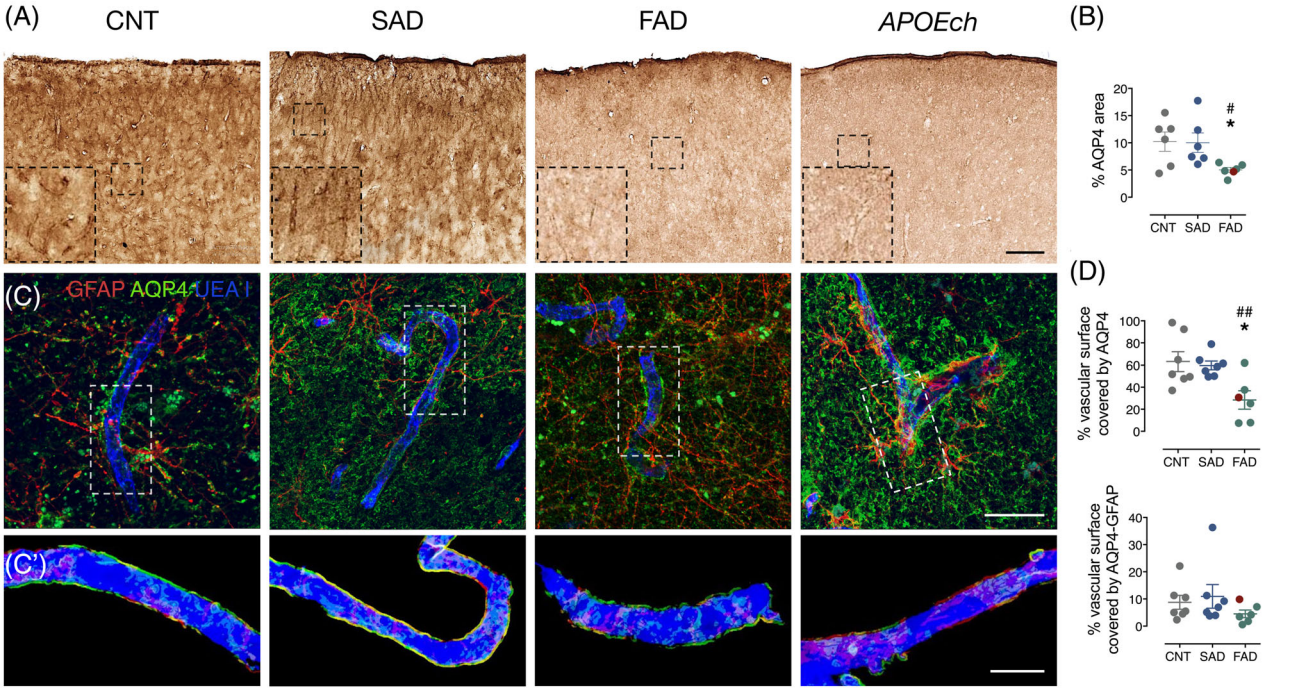


FIGURE 6 Decreased AQP4 vascular coverage in the frontal cortex of FAD and APOEch but not in SAD. (A) Representative images of layers I–III of the GM immunostained against AQP4 showing higher expression of AQP4 in CNT and SAD than in FAD and APOEch. Insets highlighting the typical vascular staining pattern found in each condition. (B) Quantification of percentage of AQP4 area in layers I–III. (C) Representative maximum intensity z-projections of confocal images in layer III of the GM showing GFAP-AQP4 astrocytes near the endothelium labeled with UEA I lectin. (C′) 3× magnified 3D reconstructions of the boxes in (C) showing the perivascular astrocytic processes present between 0 and 2 μm away from the vascular surface. The boxes in (C) are flipped 90° to the right in (C′). (D) Quantification of the percentage of vascular surface covered by AQP4 and GFAP-AQP4 astrocytic processes. Scale bars in (A): 250 μm; (B): 25 μm; (C′): 10 μm. Data are presented as mean ± SEM from CNT, $n = 7$ (C1, C2, C3, C4, C5, C6, C10); SAD, $n = 7$ (S1, S2, S3, S4, S7, S8, S9); and FAD, $n = 5$ (F1, F2, F4, F5, F6); independent experiments. The red point is the mean of three replicates of APOEch. One-way ANOVA, *t*-test, and Tukey's multiple comparison tests were performed. Comparison with CNT is denoted by asterisk (*); with SAD by number sign (#). $*^{/\#}$ indicates $p < 0.05$, and $**^{/\#\#}$ $p < 0.01$

3.7 | AQP4 vascular coverage is similarly reduced in the frontal cortex of FAD and APOEch, but not in SAD

Considering that AQP4 is the gold-standard marker for astrocyte endfeet, and its alteration has been associated with glymphatic dysfunction [103] and increased Aβ burden [104], we analyzed AQP4 expression and vascular coverage in layer III of the GM, by immunohistochemistry and 3D reconstruction, respectively. We found that AQP4 clearly delineates the vasculature in the GM of CNT and SAD (Figure 6A). Contrarily, a reduction in its expression and subtly-stained vascular structures were observed in FAD and APOEch. Consistently, quantification of percentage of AQP4 area evidenced a significant depletion in FAD and APOEch (Figure 6B). Similarly, the vascular surface covered by AQP4 significantly decreased in FAD and APOEch, and followed the same tendency in the case of double-positive AQP4-GFAP (Figure 6C,D). In contrast with the results obtained with other astrocytic markers, APOEch did not show a recovery in AQP4 expression or vessel coverage, instead, it showed a similar pattern to the rest of E280A samples that were significantly below CNT and SAD. Altogether, our results suggest that AQP4 reduction would be a hallmark of the frontal cortex of E280A, even in presence of APOEch mutation.

3.8 | The structure of the capillaries is greatly compromised in SAD and enhanced in APOEch and FAD

Considering the potential influence that different sorts of perivascular astrocytes might have on the vessels, we evaluated the 3D structure of the capillaries in layer III of the GM using confocal microscopy and 3D reconstruction. By means of vascular labeling with UEA I lectin, we found that the number of vessels per analyzed field of view significantly increased in SAD and to a lesser extent in FAD, while APOEch remained similar to CNT (Figure 7A,B). Similarly, the number of UEA I fragments and the percentage of gaps area increased in SAD but not in FAD or APOEch. In addition, the vascular diameter was meaningfully smaller in SAD than in CNT, FAD, and APOEch (Figure 7C). While the

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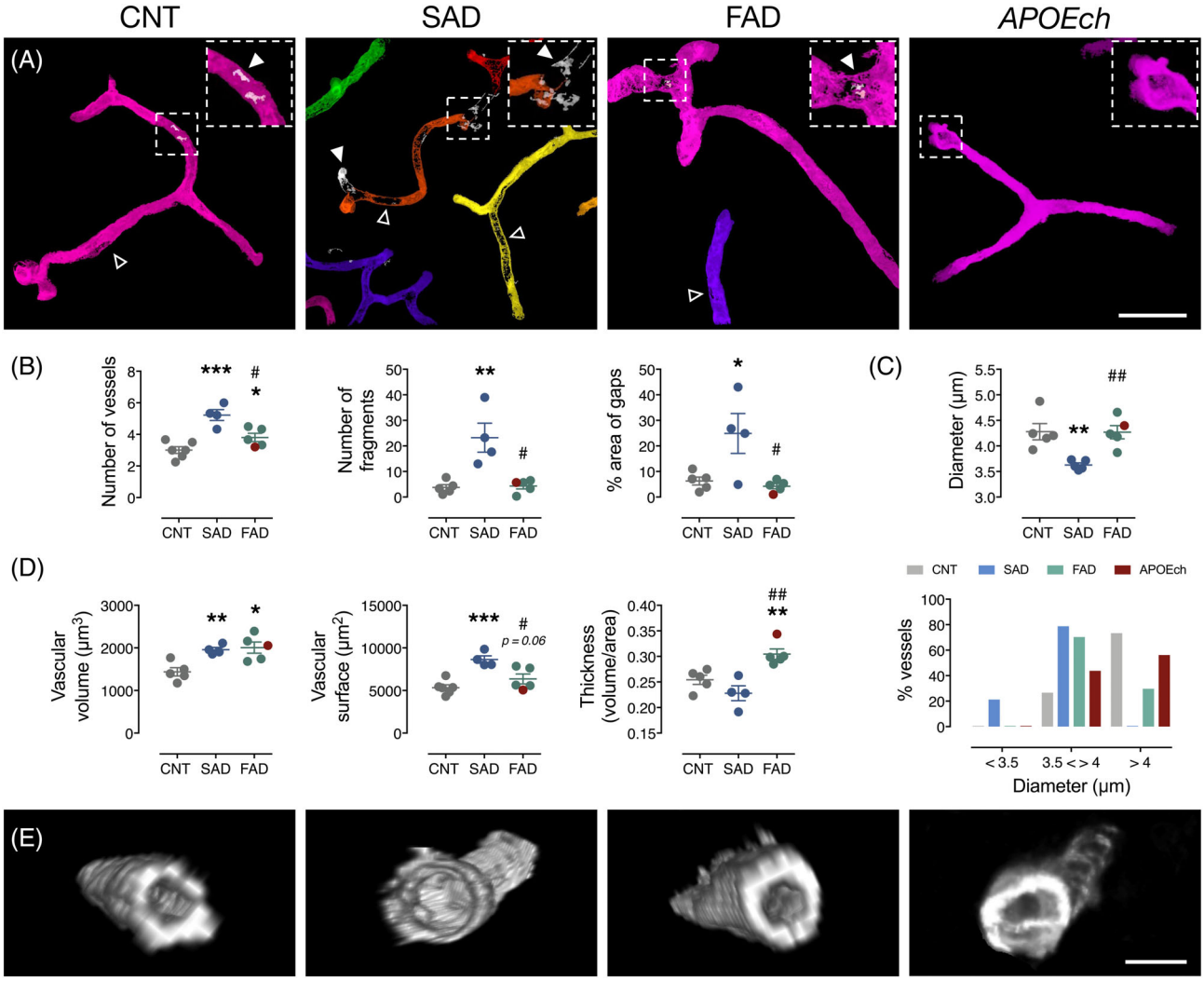


FIGURE 7 Structural deterioration of the vasculature in the frontal cortex of SAD but not in FAD nor APOEch. (A) Representative 3D reconstructions of UEA I-labeled microvasculature in layer III of the GM. Each vascular structure is different-colored. Gaps and fragments are indicated by empty and white arrowheads, respectively. Insets highlight structural damage in the capillaries. (B) Representative 3D cross-section of the microvasculature highlighting the thickness of the capillary wall. (C–E) Quantification of (C) abundance, structural integrity, (D) diameter, and (E) wall thickness of the capillaries. Scale bars in (A): 25 µm; (B): 5 µm. Insets in (A): 2× magnification. Data are presented as mean ± SEM from CNT, $n = 7$ (C1, C2, C3, C4, C5, C6, C10); SAD, $n = 5$ (S1, S2, S3, S4, S7); and FAD, $n = 5$ (F1, F2, F4, F5, F6); independent experiments. The red point is the mean of three replicates of APOEch. One-way ANOVA, *t*-test, and Tukey's multiple comparison tests were performed. Comparison with CNT is denoted by asterisk (*); with SAD by number sign (#). $^{*/\#}$ indicates $p < 0.05$, $^{**/\#\#}$ $p < 0.01$, and $^{***/\#\#\#}$ $p < 0.001$

diameter of the capillaries in SAD ranged between <3.5 and 4 µm, in CNT, FAD, and APOEch was always thicker than 3.5 µm. Indeed, a larger proportion of the capillaries in CNT and APOEch were thicker than 4 µm. As expected, considering the number of vessels found in each condition, the volume of the capillaries per field was greater in SAD and FAD than in CNT. However, no differences were found between SAD and FAD, and APOEch was close to the mean in FAD (Figure 7D). It was surprising to find that the total surface of the capillaries was only greater in SAD, while FAD and APOEch did not differ from CNT. Thus, we evaluated the volume-surface ratio as an approximation of the thickness of the vessel wall. Interestingly, FAD vessels showed higher values than SAD and CNT, suggesting thicker vascular walls, but vessels of APOEch presented the highest thickness within the E280A group (Figure 7E). These findings indicate that the number of vessels and their structural integrity are mainly altered in the frontal cortex of SAD. Outstandingly, our results also suggest a thickening of the vascular walls in the E280A group, a phenomenon enhanced in APOEch.

3.9 | Alteration of endothelial proteins in the frontal cortex of SAD and FAD but not in APOEch

As endothelial proteins support the functional properties of the brain microvasculature [11, 105], we evaluated the

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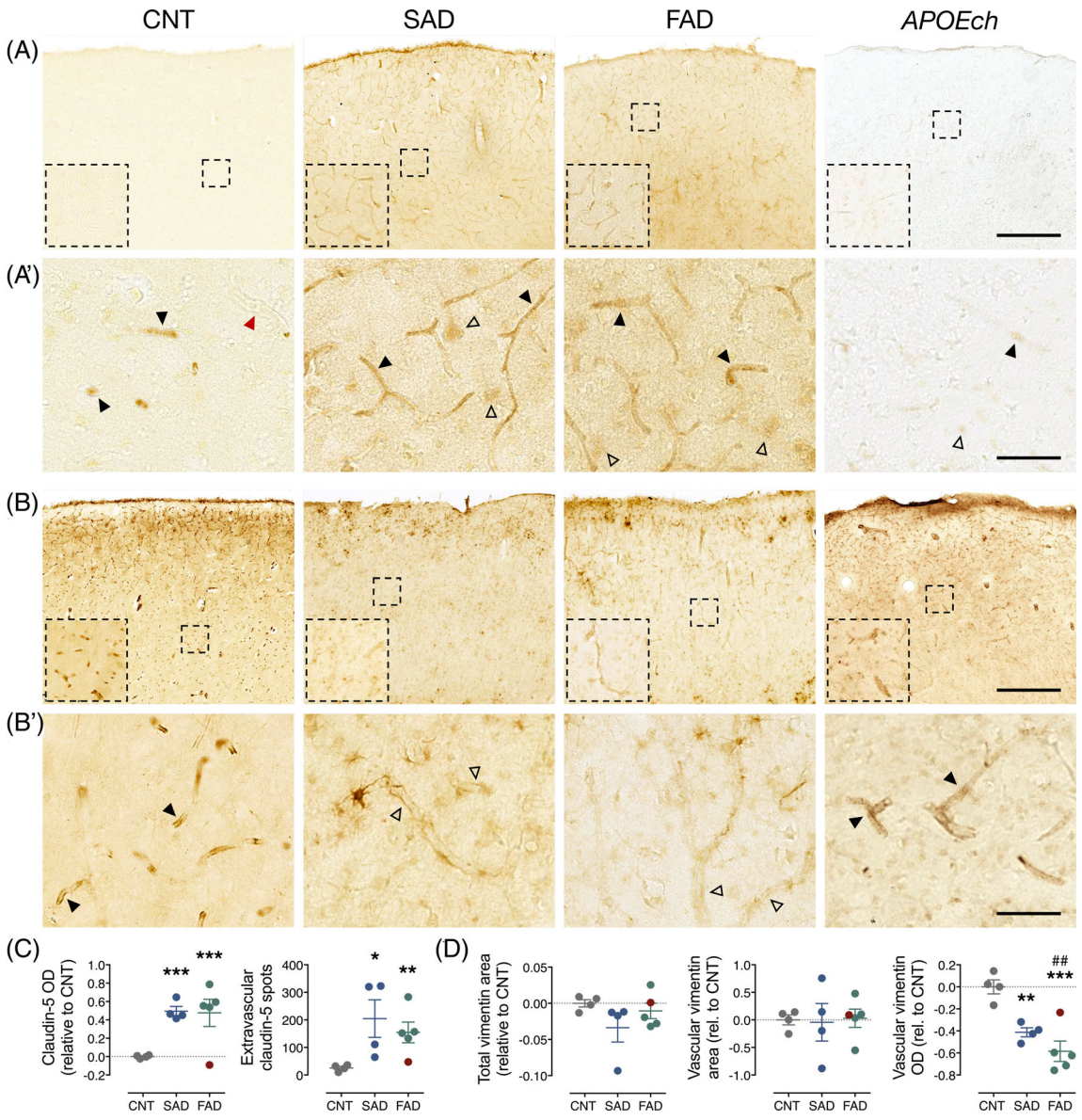


FIGURE 8 Alteration of BBB-related proteins in the frontal cortex of SAD and FAD but not in APOEch. (A–B′) Representative images of layers I–III of the GM immunostained against (A, A′) claudin-5 and (B, B′) vimentin. (A) The expression of claudin-5 increases in the microvasculature of AD but not in APOEch. (A′) Magnified images showing high expression of claudin-5 in most of the vessels of SAD and FAD. Both vascular (black arrowheads) and extra-vascular (empty arrowheads) staining was identified in AD. Vessel-like structures negative for claudin-5 are indicated by red arrowheads. (B) The expression of vascular vimentin decreases in AD but not in APOEch. (B′) Magnified images revealing similar staining patterns of vimentin between SAD and FAD but different from APOEch which resembles CNT. Vimentin inside the vessels is indicated by black arrowheads while vessels delineated by vimentin-positive astrocytic processes by empty arrowheads. Insets in A and B highlight the typical expression pattern of claudin-5 and vimentin found in each condition. (C) Quantification of (C) vascular and extravascular claudin-5, and (D) total (astrocytes plus vasculature) and vascular vimentin. OD = optical density. Scale bars in (A), (B): 500 μm; (A′), (B′): 50 μm. Insets in (A) and (B): 3× magnification. Data are presented as mean ± SEM from CNT, $n = 4$ (C1, C2, C3, C4); SAD, $n = 4$ (S1, S2, S3, S4); and FAD, $n = 4$ (F1, F2, F4, F5); independent experiments. The red point is the mean of three replicates of APOEch. One-way ANOVA, *t*-test, and Tukey's multiple comparison tests were performed. Comparison with CNT is denoted by asterisk (*); with SAD by number sign (#). $^{*/\#}$ indicates $p < 0.05$, $^{**/\#\#}$ $p < 0.01$, and $^{***/\#\#\#}$ $p < 0.001$

staining pattern of claudin-5 and vimentin in the GM. Unlike CNT and APOEch where claudin-5 was subtly detected, brain vessels in SAD and FAD displayed a remarkable high expression (Figure 8A and Figure S2D). Higher image magnification revealed increased intravascular and extravascular claudin-5 in AD compared to CNT and APOEch (Figure 8A′). Whereas almost all of the microvascular structures were claudin-5-positive in SAD and FAD, numerous capillaries did not show any staining in CNT and APOEch. These observations were quantitatively confirmed, and no differences between SAD and FAD were found in either optical density or extravascular staining (Figure 8C). Furthermore, although vimentin was typically observed throughout the GM in the vasculature

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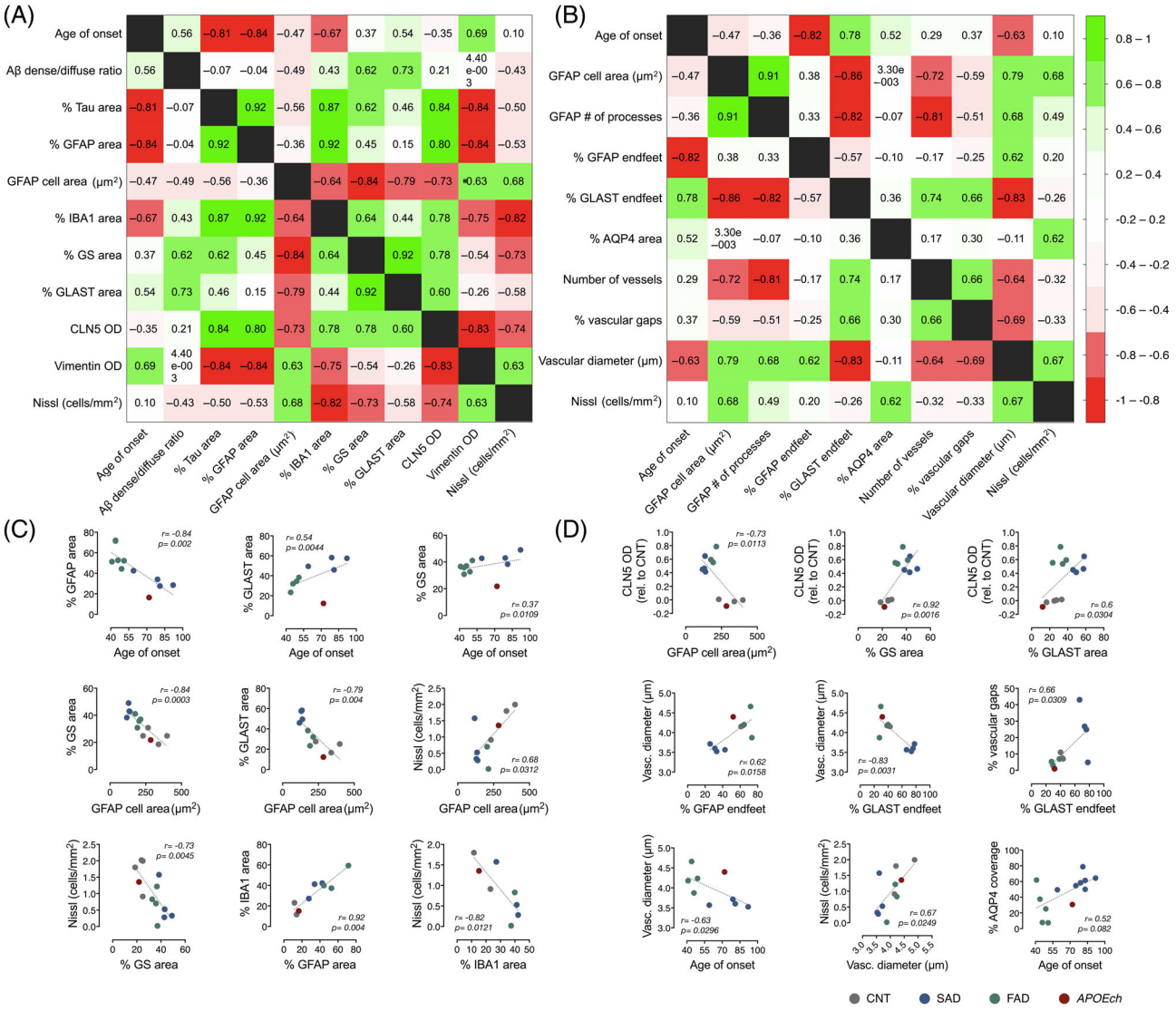


FIGURE 9 Differential alteration of functional astrocytic and vascular integrity markers in the frontal cortex of SAD, FAD, and APOEch correlate with specific reactive astroglial phenotypes. (A, B) Pearson correlation matrix showing the relationship between (A) histopathological, glial reactivity-related, astrocytic glutamate-related, and BBB-related variables; and between (B) astroglial morphology, vascular astrocytic coverage, and vascular integrity variables. (C, D) Correlation charts illustrating the segregation of SAD and FAD groups and the proximity between APOEch and CNT in variables from A and B associated to (C) neuroinflammation and astrocytic functional markers, and to (D) vascular integrity markers. Data from CNT, $n = 10$ (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10); SAD, $n = 9$ (S1, S2, S3, S4, S5, S6, S7, S8, S9); FAD, $n = 9$ (F1, F2, F3, F4, F5, F6, F7, F8, F9); and APOEch, $n = 1$; were included where available. Pearson's correlation coefficient test was performed. *r* indicates Pearson's coefficient, and *p* indicates *p* value

of CNT, it was less frequent in SAD and FAD and moderately common in APOEch (Figure 8B). Notably, the staining pattern of vimentin in CNT and APOEch was observed inside the microvascular structures, whereas in AD, the vessels were mainly outlined by astrocytic processes (Figure 8B′). While the analysis of total and vascular vimentin area did not show any difference between conditions, the optical density confirmed a significant vimentin decrease in AD, that was lower in FAD than in SAD, and similar between APOEch and CNT (Figure 8D). Therefore, the vascular decrease in vimentin detection is related to a depletion of its optical density rather than its area. In summary, the expression and distribution of vascular proteins are altered in AD but not in APOEch, suggesting disturbance of the BBB properties in AD, but comparable structural integrity in APOEch and CNT.

3.10 | Differential glial and vascular deterioration in the frontal cortex of SAD and FAD correlate with specific reactive astroglial phenotypes: APOEch gliovascular protection

To determine whether the histopathological, astroglial, microglial, gliovascular, and vascular perturbations found in the frontal cortex of SAD, FAD, and APOEch

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are interrelated, a correlation analysis was performed (Figure 9 and Figure S7). Given that age of onset differentiates FAD from SAD (Table 1), age-dependent gliovascular alterations were analyzed (Figure 9A,B). Specifically, the *age of onset* of AD negatively correlated with *p-tau area* and *percentage of GFAP* and *IBA1 area* (Figure 9A,C), and positively correlated with *vascular diameter* (Figure 9B,D), suggesting an age-dependent glial neuroinflammatory phenotype in FAD and vascular deterioration in SAD. However, changes in morphology (*astrocyte area*) and alteration of astrocytic functional proteins (GS, GLAST, and AQP4) did not show any correlation with age. Thus, although age is related to GFAP expression, it does not explain the change in astrocyte size (*GFAP cell area*), which in turn was negatively associated with modification of GS and GLAST expression, and positively correlated with cell density (*Nissl*), suggesting that the astrocyte morphology has a correlate with functional phenotypes that could impact on neurodegeneration. Likewise, specific types of astrocytes—identified according to *cell area* and GS and GLAST expression—correlate with changes in IBA1 and claudin-5, and *number of vessels*, which are also not related to age. Subsequently, we aimed to define how the major components of the GVU are associated with each other. As expected, the augment of GFAP astrocytes was directly related with the increase of reactive-like microglia, and both in turn correlated with disturbed expression of claudin-5 and vimentin. Additionally, we found a positive association between the expression of GS and GLAST. We examined the relationship between astrocytic perivascular coverage and vascular structural integrity. Surprisingly, neither a greater presence of GFAP astrocytes nor larger astrocytes correlated with AQP4. Instead, we found that alteration of *GFAP cell area* is related to altered presence of perivascular GLAST and disturbance of microvascular structure (*number of vessels* and *vascular diameter*).

Based on data from single nuclei sequencing of the frontal cortex of APOEch [65] these correlations between astrocytes and the GVU were supported by gene network analysis. Several genes belonging to gene ontology terms *maintenance of blood brain barrier* (GO:0035633) and *regulation of glial cell differentiation* (GO:0045685) were identified as differentially expressed genes in astrocytes suggesting a regulatory function on the GVU (Table S2). These genes showed to be enriched for terms related to the regulation of cell types associated with the GVU (Figure S8A), and the resulting gene network included a hub of key interconnected genes: STAT3, CLU, TGFB1, MAPK8, VEGFA, and NFKB1. CLU, APOE, GFAP, and GPC6 were identified as differentially expressed in the frontal cortex of APOEch (Figure S8B), suggesting a possible effect on GVU homeostasis.

Furthermore, we identified two types of Aβ plaques morphologies with differential abundance in SAD and E280A (Figure S1). No association was found between the deposition of Aβ (*% Aβ area*) and GFAP increase or vascular injury (Figure S7). In contrast, the proportion of dense or diffuse plaques (*Aβ dense/diffuse ratio* and *% Aβ diffuse area*) correlate with p-tau, and astrocyte and vascular alterations. Moreover, the *duration of the disease* was only positively correlated with the amount of p-tau, and as expected, the age of death was inversely correlated with astrocytic size, gliovascular coverage, and vascular diameter. Outstandingly, neither the age (of onset or death), nor the load of Aβ and p-tau correlated with the cell density (*Nissl*) of the frontal cortex. Instead, it was negatively affected by the increase of ameboid-like microglia, the expression of GS, and by vascular deterioration variables such as the alteration of claudin-5 and vimentin, as well as by capillary diameter. Lastly, plots of the cellular variables that showed the strongest correlations revealed the segregation of SAD and FAD groups and exposed the proximity of APOEch to CNT, especially in analyses related to astroglial and microglial reactivity, astrocytic glutamate-related proteins, and vascular integrity (Figure 9C,D). Altogether, these analyses support the idea that the frontal cortex in SAD, FAD, and APOEch is influenced by specific astroglial phenotypes that differently contribute to the disturbance—or maintenance—of glutamate homeostasis, and to the structural and functional integrity of the brain vasculature.

4 | DISCUSSION

In the present study, we found that specific astrocytic phenotypes coupled to the GVU define the structural integrity of the microvasculature in the frontal cortex of SAD, FAD, and APOEch. In particular, while astrodegeneration, alterations related to astrocytic glutamate-regulatory mechanisms, vascular deterioration, and disturbance of BBB-related proteins are prominent in SAD; FAD is characterized by hyperreactive-like glia; and APOEch displays the mildest astrocytic and vascular alterations despite having the highest burden of Aβ. Additionally, we determine that vessels less covered and primarily associated to processes positive for glutamate-regulatory proteins characterize frontal cortex of SAD; while FAD is distinguished by vessels associated to GFAP processes, and devoid of AQP4. We report that the effects of the ApoE3ch mutation in an E280A carrier might be related to the promotion of astrocytic and gliovascular homeostatisis despite the massive load of Aβ.

Our observations of different staining patterns of Aβ and p-tau in the GM of SAD and FAD are in accordance with previous reports [75]. However, we describe here greater deposition of both proteins in the upper layers of the GM than in the deepest, and demonstrate that APOEch displays the highest burden and broadest distribution of Aβ and the lowest levels of p-tau among the studied conditions [65]. The lower cell densities in AD could be related to a greater presence of Aβ and p-tau or even to age, however, the greatest load of Aβ in APOEch

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cast doubt on the direct relationship that has usually been suggested between Aβ and neurodegeneration, and instead highlight the potential relevance of other elements such as those of GVU.

Vimentin and GFAP link astrocyte structure and gliovascular function [90, 92], and are associated with cytoplasmic proteins [92] participating in signal transduction pathways [44, 45, 106, 107]. Therefore, fine-tuned modification of astrocytic GFAP and vimentin expression is critical for the execution of the astrocyte reactive program [28, 71, 91]. Our observations of differential vimentin and GFAP expression between SAD, FAD, and APOEch suggest that astrocytes undergo different reactive responses in the frontal cortex of each condition. In fact, it has been reported that GFAP increases in parallel with Braak stages in AD and correlates with greater astrocyte reactivity [41]. Hence, it is remarkable that APOEch displayed the lowest levels of GFAP and its staining pattern was similar to that of CNT, which presented characteristics coherent with previous descriptions of healthy cortex [91, 106, 107]. Thereby, a higher GFAP in FAD, followed by SAD is in accordance with Braak staging, but it is contradictory for APOEch which presented one of the highest Braak scores.

GFAP immunolabeling clearly delineates cell morphology [92]. Typical human astrocytes display a radial stellate morphology [38, 108], however, alteration of cell shape is one of the changes associated with astroglial reactivity and modification of vimentin and GFAP [15, 28, 42, 43, 91]. Two broad astrocytic morphological categories can be distinguished. Atrophied astrocytes are characterized by decreased cell size and reduction of complexity [45–47]. Under this context, astroglial death increases [28, 47]. In contrast, hypertrophied astrocytes display thicker intermediate filament bundles, highly upregulate GFAP [45, 109] and are associated with neuroprotection and postlesion regeneration [28, 45–47]. Nonetheless, under prolonged stress it can result in chronical inflammation [20, 36, 105, 109]. The astrocyte morphology identified in SAD resembles that of degenerated astrocytes, while that in FAD is more related to a hyperreactive phenotype. Outstandingly, astrocytes in APOEch did not show remarkable differences with those of CNT, suggesting similar functional properties of healthy astrocytes. Astroglial proliferation has also been related with GFAP upregulation [71, 91, 110]. Accordingly, our group has shown that postmortem cultured astrocytes from the frontal cortex of FAD display greater proliferative capacity than those of SAD [111]. We have also shown that blood serum-derived extracellular vesicles from FAD are associated with inflammatory-related factors to a greater extent than those from SAD [73]. Typically, activated microglia displays ameboid morphologies [112] and induces astrocytic activation [106, 113, 114]. Thus, although increase of amoeboid-like microglia occurred in both SAD and FAD—as previously described in AD [106, 114, 115]—the higher expression of IBA1 in FAD is consistent with the hyperreactive-like astrocytic phenotype. Meanwhile, APOEch exhibited the lowest IBA1 levels and atrophied-like microglia, also correlating with its low GFAP levels and the intermediate reactive-like astrocyte morphology. Together, these reports support the potential relationship between the differential reactive GFAP patterns identified in SAD, FAD, and APOEch with specific astrocytic phenotypes.

It has been described that GS decreases in aging [47], AD [112], and reactive astrocytes [94]. However, it has also been shown that GS can be upregulated by astrocytes [95] and expressed by neurons [24] in response to aging and AD. Whatever the case, altered GS expression is associated with a cytotoxic microenvironment because of imbalance in excitatory-inhibitory neurotransmission. Similarly, prior reports have shown that Aβ [113] and inflammation [96] decrease GLAST expression and function [114], weaken glutamate clearance [115, 116], and favor excitotoxicity [25, 28]. Although alterations in the expression of both GS and GLAST are described in AD, it was surprising to find slight differences between SAD and FAD, and more conspicuous variations when compared to APOEch. For instance, previous reports have shown that atrophied astrocytes fail to maintain the glutamate-glutamine shuttle [28, 47] and display insufficient glutamate clearance [43] because of reduced synaptic coverage [45–47], contributing to synaptic extinction and cognitive deficiency [43]. Conversely, hyperactivated astrocytes and microglia, support excitotoxicity through the exacerbated release of glutamate in response to Aβ [37, 38, 40, 43, 50, 51, 117, 118], and even GABA-producing astrocytes are commonly found in AD [42, 97, 119]. Hence, it is plausible to argue a correlation between the different GFAP astroglial phenotypes observed in the frontal cortex of SAD, FAD, and APOEch, with the differential alteration of astrocytic glutamate-related proteins. This is supported by the fact that adequate GFAP expression is necessary for retaining GLAST in the plasma membrane [36]. Indeed, although the subcellular localization of GLAST is typically perisynaptic, it is highly mobile within the astrocyte membrane influenced by neuronal and glial activity [120]. Therefore, the GLAST aggregates observed in AD and APOEch—whose amounts are comparable with the GFAP levels of each group (the greatest in FAD, followed by SAD and lastly by APOEch)—might have resulted from its redistribution, presumably responding to glutamate-induced stress and as a consequence of the increase of astroglial reactivity. Our results show that the main astrocytic glutamate-regulatory proteins GS and GLAST are perturbed in the frontal cortex of SAD, FAD, and APOEch, which might reflect a link with particular reactive and functional astrocytic phenotypes.

Astrocyte reactivity is tightly linked to endothelial [50], which are in turn associated with loss of vascular structural integrity and opening of the BBB [15, 28, 38,

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45, 107, 109, 121]. Both proinflammatory molecules and glutamate have been broadly shown to impair the structure and function of brain endothelium [50, 51]. Thereby, altered expression of GFAP, GS, and GLAST in the frontal cortex of AD and APOEch might imply modification of the interactions between astrocytes and the associated vessels. Specifically, the fact that microvessels were majorly associated with astrocytic perivascular processes expressing GLAST in SAD, and GFAP in FAD, suggests that glutamate may play an important role in the impairment of the GVU in SAD, whereas inflammatory activity might be the predominant mechanism in FAD. Coherently, our analyses revealed differential levels of vascular deterioration between the studied conditions, such that the structure of the capillaries was markedly compromised in SAD, compared to FAD and APOEch that showed structurally strengthened vessels. It is possible that factors such as CAA (shown in Table S1) explain these differences. However, as CNT, FAD, and APOEch showed the lowest vascular deterioration and yet showed completely diverse CAA scores, we are inclined to hypothesize the existence of a relationship between the perivascular component and the structural integrity of the capillaries. In this direction, it has been shown that altered expression of GS is associated with changes in endothelial cell migration and vessel sprouting [122], while ablation of reactive astrocytes prevents the reconstruction of injured BBB [28, 38, 45]. In addition, reduced vascular diameters and increased capillary densities [123] have been documented in aged individuals and AD [124]. Loss of perivascular AQP4 has been associated with glymphatic dysfunction and accumulation of Aβ [104], which supports our findings of AQP4 decrease and high Aβ loads in the frontal cortex of FAD and APOEch. These results were also consistent with previous reports of reduced AQP4 perivascular in E280A carriers [66]. Homeostatic GVU regulation in APOEch seems to be determined by CLU and APOE gene expression, possibly via cholesterol efflux and lipid regulation on the BBB [125].

Moreover, both endothelial tight junction claudin-5 and vimentin influence the functional stability of the cerebral vasculature [11, 105]. In particular, claudin-5 is crucial for the selective permeability of the BBB [10], and neuroinflammation induces its redistribution favoring hypervascularization and BBB leakage [126]. While the implications of endothelial vimentin have not been addressed in AD, its depletion in other models results in leukocyte extravasation and decreased blood flow [11]. Although we found that both endothelial proteins are altered in SAD and FAD, we were unable to demonstrate a decrease in claudin-5 in AD, as previously reported [127]. Instead, we observed increased and altered expression pattern that might have resulted from its redistribution. Remarkably, we had previously shown that potentially claudin-5-positive extracellular vesicles increase near the vessels of the frontal cortex in a manner that differentiates SAD from FAD [73]. Altered expression of claudin-5 and vimentin might reflect functional impairments in the vasculature of AD. Surprisingly, both the vascular structure, as well as the expression of claudin-5 and vimentin in APOEch were similar to those of CNT. These observations suggest that specific astrocyte populations surrounding the brain vessels might underlie some of the structural differences that characterize the vasculature of each condition.

ApoE binding to heparan sulfate proteoglycans (HSPGs) has been suggested to promote Aβ aggregation and neuronal uptake of extracellular p-tau [62]. The protective effect of ApoE3ch is related to a lower affinity to bind HSPGs or other ligands [62]. Notably, ApoE is also involved in the modulation of glutamate transport and metabolism [128], and astrocytes and microglia are the main ApoE-producers in the brain [64]. APOEch showed that the astrocytes from the frontal cortex display a homeostatic transcriptional profile contrary to other studied brain regions [65]. In this study, ApoE3ch was found to be a glioprotective factor associated to a greater preservation of the GVU structure and the vascular integrity. Despite our findings are restricted to the frontal cortex and further investigations assessing these phenomena in other cerebral areas are necessary. Other genetic variants that have also been identified as modifiers or protective against FAD, delay AD-onset and are related to astrocyte genes, ubiquitin-proteasome function, and p-tau degradation [61, 126, 129]. Considering that ApoE is involved in p-tau spreading and regulation, ApoE and astrocytes seem to be convergence factors in FAD protective variants.

Altogether, the human frontal cortex of SAD, FAD, and APOEch is characterized by specific astrocyte phenotypes which define the integrity of GVU. Astrocytes may play a role in the mechanisms of protection conferred by APOEch. This study provides new insights into the potential relevance of the gliovasculature in the development and protection of AD.

AUTHOR CONTRIBUTIONS

Rafael Posada-Duque was involved in conceptualization (lead), funding acquisition (lead), and project administration (lead). Julián Henao-Restrepo performed the experiments and analyzed the data. Julián Henao-Restrepo, Gloria Patricia Cardona-Gómez, and Rafael Posada-Duque designed the experiments. Gloria Patricia Cardona-Gómez, and Johanna Gutiérrez-Vargas supported in conceptualization, funding acquisition, and project administration. Johana Gomez, Andrés Villegas, Joseph F. Arboleda-Velásquez, and Francisco Lopera were involved in sampling and diagnosis of human tissue. Julián Henao-Restrepo standardized image processing and analyses. Carolina López-Murillo was involved in microglia cell morphology and AQP4 processing and quantification. Pablo Valderrama-Carmona was involved in AQP4-GFAP-UEA1 imaging, processing, and quantification. Natalia Orozco-Santa was involved in GLAST-

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GFAP-UEA1 imaging, processing and quantification. Renato Moraga was involved in vimentin, claudin-5, Aβ, Nissl processing and quantification. Jessica Lisa Littau analyzed the gene network. Diego Sepulveda-Falla helped in morphological comparison of APOEch astrocytes and microglia in frontal, temporal, parietal, and occipital cortices. Renato Moraga, Jorge Toledo, and Steffen Härtel were involved in acquisition, processing, and quantification of images from spinning confocal microscopy and whole slide microscopy. Julián Henao-Restrepo and Rafael Posada-Duque wrote the manuscript. All authors contributed to the article drafting and approved the submitted version.

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CONFLICT OF INTEREST

The authors declare that this research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

DATA AVAILABILITY STATEMENT

The original contributions presented in the study are included in the article/supplementary material.

ETHICS STATEMENT

The procedures using human brains were approved by Record 119, Agosto 2018-Sede de Investigaciones Universitarias, Universidad de Antioquia. Written informed consent was obtained from the individuals for the publication of any potentially identifiable images or data included in this article.

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